

Adaptive M5 Volume-Confirmed Reversal: Detailed Research Paper

RegimeForgeEA

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Abstract

This paper documents an M5 long reversal candidate and a subsequent risk-adaptive overlay. The base rule was selected with a 2021-2023 training set, validated on 2024, and evaluated once on a 2025 holdout using public PAXGUSD data as a gold-linked proxy. The adaptive overlay adds a completed H1 trend filter, a daily realized-loss lock, and a loss-streak cooldown. It reduces the observed 2025 drawdown, but it was designed after inspecting the original holdout result; consequently it is exploratory and requires a new independent broker-native XAUUSD test before any deployment decision.

1. Research question

The base candidate has positive proxy-data performance but an unacceptable 2025 drawdown at the original 0.10 lot reference size. The purpose of the adaptive overlay is to avoid mean-reversion entries during broader downtrends and to limit repeated losses during adverse intraday conditions, without modifying the underlying alpha signal.

2. Data and execution assumptions

The research uses weekday-only PAXGUSD five-minute bars from 2021 through 2025. PAXGUSD is a public gold-linked proxy and is not an XAUUSD broker feed. Aggregate-trade-derived volume is used for the proxy factor. The backtest enters at the next M5 open, charges a fixed 0.35 USD round-trip spread, exits at a fixed horizon, and prohibits position overlap. Dollar PnL is expressed with a 0.10 lot and 100 oz contract reference solely for comparability.

The following are outside scope: broker Bid/Ask history, slippage, commissions, swaps, fill latency, stop-level rules, symbol suffixes, weekend gaps, and order rejections. Any performance result in this paper is therefore research evidence rather than an XAUUSD or live-trading claim.

3. Base factor specification

Let C_t and V_t denote the close and volume of the closed M5 bar t .

$$r_3(t) = \frac{C_t}{C_{t-3}} - 1$$

Let $q_{20}(t)$ be the 20th percentile of the preceding 5,760 three-bar returns (20 trading days), calculated with a one-bar shift. The shift ensures that the threshold at t excludes the current signal return.

$$q_{20}(t) = Q_{0.20}(\{r_3(s) : s < t\})$$

Define the volume ratio:

$$VR(t) = \frac{V_t}{SMA_{60}(V)_t}$$

The base long signal is:

$$r_3(t) \leq q_{20}(t) \quad \wedge \quad VR(t) \geq 1.5$$

At the signal close, the strategy submits a long order at the next M5 opening price. It exits after 24 M5 bars (120 minutes). Entry and exit each include half of the fixed 0.35 USD round-trip spread.

4. Base-rule selection protocol

The original grid used direct, non-overlapping order-level simulations. It considered order-flow, return reversal, break-out reversal, and volume-confirmed reversal families; long and short directions; Asia, London, New York, and all UTC sessions; and holding periods from 5 to 120 minutes.

The training candidate gate required at least 100 trades, profit factor (PF) at least 1.10, and mean net return at least 0.50 bps. The selected candidate then required at least 30 validation trades, PF at least 1.05, and mean net return at least 0.25 bps. The 2025 sample was inspected only after the base candidate passed 2024 validation.

5. Drawdown diagnosis

The base rule is a long mean-reversion strategy. Its adverse periods are concentrated in persistent downtrends and elevated volatility, where repeated oversold readings do not revert promptly. In the 2025 proxy holdout, the worst monthly results occurred in May, October, and December, and the maximum drawdown trough occurred on 21 November 2025. The 0.10 lot reference drawdown was 4,927.40 USD, compared with 6,707.20 USD total holdout PnL.

6. Adaptive overlay

The overlay does not change the base alpha condition. It controls when new signals may be traded.

6.1 Completed H1 trend filter

Only trade when the prior completed H1 bar has a positive EMA trend:

$$EMA_{20}^{H1}(t-1) > EMA_{50}^{H1}(t-1)$$

The completed-bar shift prevents use of an unfinished H1 candle.

6.2 Daily realized-loss lock

Let B_d be the account balance at the start of broker day d , and let R_d be realized PnL accumulated during that day. Reject new entries when:

$$R_d \leq -0.02B_d$$

The lock resets on the next broker day. It does not forcibly close an existing position; it only stops subsequent entries.

6.3 Loss-streak cooldown

After four consecutive losing exits, do not open a new position for 12 M5 bars (one hour). A profitable exit resets the loss counter. The cooldown is intended to avoid clustered entries when the short-horizon reversal premise is failing.

7. Results

Sample	Base PF	Adaptive PF	Base net PnL	Adaptive net PnL	Base max DD	Adaptive max DD
Training 2021-2023	1.156	1.162	11,644.50	6,063.90	-2,111.00	-2,251.00
Validation 2024	1.257	1.334	7,832.00	5,565.00	-1,847.00	-946.00
Holdout 2025	1.119	1.221	6,707.20	7,151.00	-4,927.40	-3,134.50

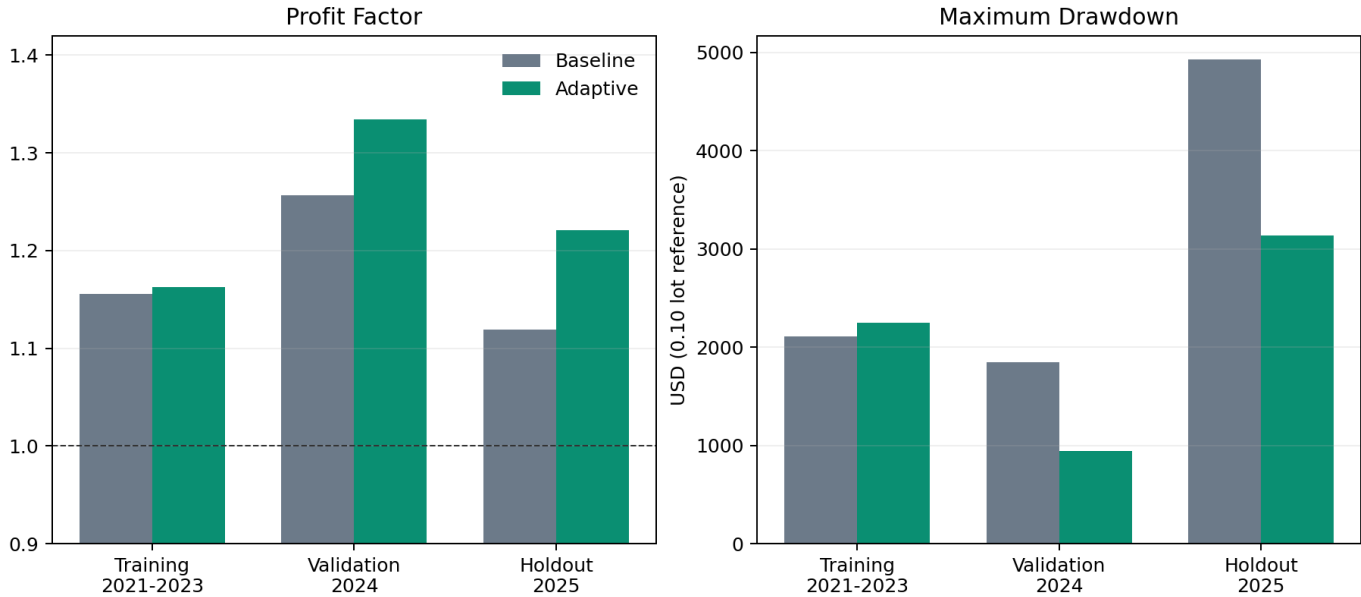


Figure 1: Base and adaptive comparison

The overlay improves PF in every displayed sample. However, training drawdown is slightly worse, so the overlay must not be presented as universally superior. It is a risk-oriented adaptation that materially improves the observed 2024 and 2025 drawdowns, not an optimized global solution.

8. Position sizing interpretation

The 0.10 lot reference is too aggressive for a 10,000 USD account. With a linear approximation, the adaptive 2025 maximum drawdown of 3,134.50 USD at 0.10 lot becomes approximately 626.90 USD at 0.02 lot, or about 6.3% of a 10,000 USD account. This is an illustration, not a risk guarantee: real XAUUSD spread, slippage, contract size, and gap behavior can break linear scaling.

The provided EA therefore defaults to `InpFixedLots=0.02`. The professional tester should use the broker's contract specification and an account-level risk budget rather than copying this illustration mechanically.

9. MQL5 implementation mapping

`Experts/RegimeForgeVolumeReversalEA.mq5` implements the following semantics:

1. Evaluate the volume-confirmed reversal signal on the last completed M5 bar.
2. Calculate the threshold from prior bars only.
3. Require the previous completed H1 EMA20/EMA50 trend condition.
4. Block new entries after the daily realized-loss limit or a loss-streak cooldown.
5. Use the EA magic number to isolate positions and exit the managed position after the configured holding interval.

The EA intentionally defaults to `InpEnableNewEntries=false`. A tester must explicitly enable entries in MT5 Strategy Tester.

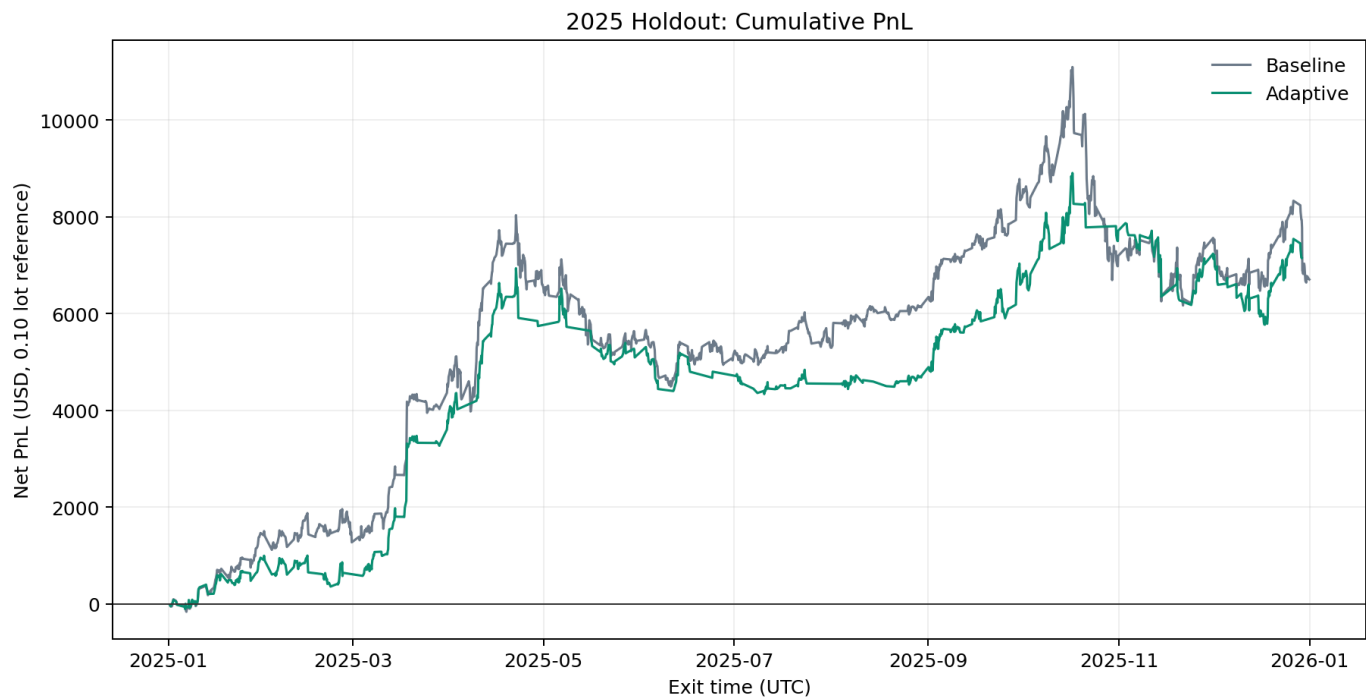


Figure 2: 2025 holdout equity comparison

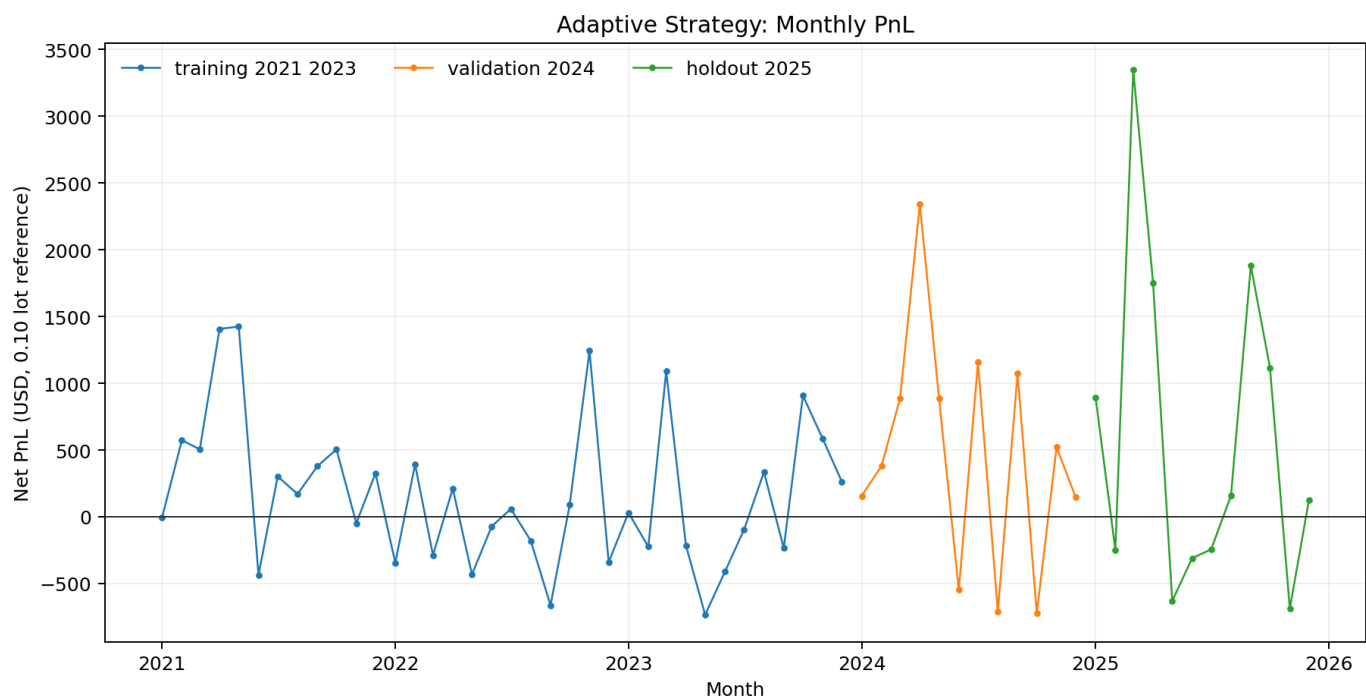


Figure 3: Adaptive monthly PnL

10. Professional XAUUSD test protocol

The professional tester should use real ticks or the best available broker Bid/Ask model, fixed symbol specifications, and a full commission/swap setup. At minimum, record:

1. PF, expectancy, maximum drawdown, recovery factor, and trade count.
2. Sensitivity to spread expansion, slippage, and execution delay.
3. Performance across at least three non-overlapping calendar periods.
4. Results with the base rule and adaptive overlay separately.
5. Risk at 0.01, 0.02, and a broker-appropriate risk-sized volume.
6. Behaviour around news, rollover, and sessions with poor liquidity.

The adaptive configuration should be considered accepted only if its XAUUSD result remains positive after realistic costs and its drawdown stays inside the tester-defined risk budget.

11. Reproduction

```
python scripts/research_order_flow_absorption.py \  
  data/derived/PAXGUSD_5m_2021_2025_weekdays.csv \  
  data/derived/PAXGUSD_order_flow_5m_2021_2025.csv \  
  --factor volume_return_3_reversal --bar-minutes 5 --hold-bars 24 \  
  --side long --session-hours 0-23 --use-higher-trend-filter \  
  --max-daily-loss-percent 2 --initial-equity 10000 \  
  --max-consecutive-losses 4 --cooldown-bars 12 \  
  --trades outputs/m5_adaptive_volume_reversal_trades.csv \  
  --report reports/M5_Adaptive_Volume_Reversal_Backtest.md
```

```
python scripts/generate_adaptive_factor_paper_charts.py \  
  outputs/m5_volume_reversal_holdout_trades.csv \  
  outputs/m5_adaptive_volume_reversal_trades.csv \  
  --assets reports/assets
```

12. Conclusion

The base M5 volume-confirmed reversal factor is a credible proxy-data research candidate with positive training, validation, and original holdout results. The adaptive overlay improves observed recent-sample risk but is exploratory because it was formulated after inspecting the original holdout drawdown. Broker-native XAUUSD Bid/Ask testing is the next decision gate; no live-profit conclusion is warranted before that test is complete.